

Tensor Network Solver within a Finite Volume Framework for the 2D Convecting Taylor-Green Vortex

2026 Global Quantum + AI Challenge - Phase 1 Concept Proposal

Problem statement: Airbus, 'Quantum Solvers: Enhancing Predictive Aerodynamic Modeling Capabilities'

Public code repository: <https://github.com/theaiwillwin/airbus-octonion-tgv-solver>

Experimental research output - not a validated engineering deliverable.

1. Problem Framing

Accurately predicting aerodynamic flows requires solving PDEs at HPC scale, and current methods face severe scalability limits: the memory footprint of time-resolved flow solutions grows as $O(n^3)$ with resolution, and expensive wind-tunnel testing remains necessary at demanding operating conditions. Airbus seeks more efficient PDE solvers. We address the prescribed 2D Convecting Taylor-Green Vortex (TGV) because its exact analytical solution enables rigorous, unambiguous measurement of error, runtime, and memory - the three axes on which any claimed advantage must be judged.

Why a quantum-inspired method offers credible advantage here: time-resolved flow solutions are dominated by a modest number of coherent structures, so their space-time tensors admit low-rank tensor-network representations whose storage scales with flow complexity rather than grid resolution. We demonstrate this today, classically, with measured numbers through a 512^2 grid - and the same representation maps directly onto quantum states whose qubit count grows only logarithmically with resolution, giving a concrete, honest bridge to fault-tolerant hardware.

2. Technical Approach

Paradigm: quantum-inspired (Tensor Network within a Finite Volume framework - the second track explicitly permitted by the problem statement), with a mapped pathway to gate-based fault-tolerant hardware. No quantum hardware is used or claimed in Phase 1.

Classical baseline (implemented and verified): a cell-centered finite-volume solver on the periodic $[0, 2\pi]^2$ domain - face-based convective and diffusive fluxes, RK2 time stepping with CFL control, and FFT (spectral) pressure projection enforcing incompressibility. The benchmark truth is the exact analytical convecting TGV solution; no external CFD data is required.

Tensor-network layer (implemented and measured): the full space-time (time, x, y) solution tensor is stored in Tucker-decomposed form (TensorLy) - a tree tensor network. Per-mode ranks are selected automatically by a novel octonionic control layer: local flow states (velocity, vorticity, velocity gradients) are embedded as 8-component octonions over Fano-plane structure constants, and the non-associative associator norm $|| (a*b)*c - a*(b*c) ||$ acts as a relational torsion diagnostic. High associator statistics indicate complex local flow relationships and allocate more rank; the policy is grid-independent, so rank tracks flow physics rather than resolution.

Quantum pathway: the Tucker/TTN factorization maps onto quantum state preparation with $\text{ceil}(\log_2 n)$ qubits per mode - approximately 46 logical qubits at 64^2 and 57 at 512^2 using our measured ranks (Appendix B). Qubit count grows logarithmically with grid size while classical storage grows linearly: a credible, quantified route to quantum memory advantage under fault-tolerant hardware, stated as a mapping rather than a result.

3. Feasibility and Resource Requirements

The pipeline already runs end-to-end on a standard Windows CPU (NumPy, TensorLy, pandas, matplotlib, pytest; no GPU, cluster, or QPU). All results in this proposal are measured from the public repository above; the largest case (512^2 , 4.18 GB dense trajectory) needs ~16 GB RAM and ~20 minutes. Phase 2 needs modest resources: a multi-core CPU node (or small cluster) for 3D and higher-Re extensions, and optional QPU access only for small-scale state-preparation demonstrations.

Honest constraints: (a) the TGV is spectrally narrow and favorable to low-rank storage - our perturbed-flow study measures how the advantage degrades (gracefully) as mode content grows; (b) the reduced-order-model speedup is currently an operation-count estimate, not a wall-clock measurement; (c) current quantum devices cannot execute the required state-preparation circuits - the quantum pathway is a resource-quantified mapping,

not a hardware claim.

4. Expected Impact

Phase 1 already delivers measured results rather than promises; a successful Phase 2 PoC converts the measured memory advantage into a time-to-solution advantage and validates it on harder flows. All numbers below regenerate from documented scripts.

Measured result 1 - verified solver: grid-convergence order 1.99 (design 2), divergence $\sim 1e-7$, stable across $Re = 10-2000$ (20x the mandatory $Re = 100$), 48/48 automated tests passing.

Measured result 2 - tensor-network compression, every row measured:

Grid	Dense	Tucker (guided)	Ratio	Error
64^2	2.1 MB	0.27 MB	7.8x	$1.1e-15$
128^2	32.8 MB	0.32 MB	102x	$1.2e-15$
256^2	523 MB	0.52 MB	1010x	$1.4e-15$
512^2	4.18 GB	0.86 MB	4880x	$1.3e-15$

Guided ranks saturate at (14, 29, 29) from 128^2 onward - rank tracks flow complexity, not resolution. Our pre-run projection for 512^2 (4887x, from measured rank saturation) matched the subsequent direct measurement (4880x) within 0.15%, validating the projection methodology itself. Dense storage grows $O(n^3)$; compressed storage barely grows.

Measured result 3 - the octonion-associator guidance grows with flow complexity: at matched reconstruction error the guided asymmetric rank allocation beats the best uniform rank by 1.22x on the smooth TGV, rising to 1.73x on a multi-mode perturbed TGV (10% divergence-free perturbation; 1.67x at 30%). The advantage increases in exactly the direction that matters for industrial flows.

Honest ablation - we ran the control a reviewer would run: associator-adaptive CFL steering shows $\sim 10\%$ lower L2 error, but a uniform CFL reduction without octonions reproduces it at $Re \geq 500$ at $\sim 60x$ lower runtime. We do not claim a net steering advantage on this benchmark; the associator's measured contribution is rank selection.

Quantitative Phase 2 targets: (i) measured wall-clock ROM speedup $\geq 2x$ via DEIM hyper-reduction; (ii) guided-vs-POD memory advantage at matched error on multi-mode flows; (iii) compression $\geq 100x$ at $\leq 1e-6$ error on a 3D TGV at 128^3 ; (iv) small-scale quantum state-preparation demo of a rank-truncated factor on available hardware.

5. Validation Plan

Every claim is validated against the exact analytical TGV solution and a verified classical baseline. Recorded metrics: time-to-solution, memory (dense and compressed), L2 velocity error, kinetic-energy decay, divergence norm, compression ratio, reconstruction error, and associator statistics. The Phase 1 methodology - which Phase 2 will extend - is: (1) verify the solver (grid-convergence order, divergence control); (2) measure, never estimate, wherever possible; (3) run error-matched head-to-head comparisons against the strongest naive baseline; (4) run the control experiment for every claimed advantage and report ablations that fail. Phase 2 success criteria are the four quantitative targets in Section 4, evaluated on the same measured basis, plus independent reproduction from the public repository. Mandatory cases $Re = 10$ and $Re = 100$ are complete and extended to $Re = 2000$ with grid refinement 32^2-512^2 ; 48 automated tests guard every module.

6. Hybrid / Cross-Domain Integration

The design is hybrid by construction: a classical FV solver produces physics; a tensor-network layer compresses and (in Phase 2) evolves it; the octonionic associator layer supplies geometry-aware control signals; and the TTN factorization is the hand-off point to quantum hardware, where state preparation replaces classical factor storage. Modules are cleanly separated (solver, octonion algebra, diagnostics, compression, benchmarking), so the natural aerodynamic extensions - multi-mode initial conditions, 3D TGV, higher Re - reuse the same algebra and compression pipeline unchanged. Within Airbus workflows, compressed trajectory storage integrates directly with existing HPC post-processing: 4.18 GB $\rightarrow$ 0.86 MB per trajectory changes what can be archived,

transmitted, and interactively queried from a full unsteady simulation campaign.

7. Team Capability

This submission demonstrates capability by artifact rather than resume: a working, verified solver; a novel octonionic control layer implemented from Fano-plane structure constants (exact associator, no proxy heuristics); measured results through 512^2 with machine-precision reconstruction; control experiments and honest negative results; 48 automated tests; and full public reproducibility. The working style Phase 2 requires - measure, verify, report the failures alongside the wins - is already evidenced throughout this proposal and its repository.

Appendix (supplementary material, 3 pages max)

A. Reproducibility and Rank-Selection Policy

Public repository (all code, tests, configs, results): <https://github.com/theaiwillwin/airbus-octonion-tgv-solver>

```
python -m pytest # 48/48
python scripts/run_reynolds_sweep.py --config configs/reynolds_sweep.yaml
python scripts/run_advantage_study.py
python scripts/run_scalability_projection.py --re 100 --grids 32 64 128 256
python scripts/run_perturbed_advantage_study.py --re 100 --nx 64
```

Rank policy: associator_guided_ranks uses a grid-independent cap (min 4, max 32). An earlier internal draft used a grid-dependent cap, which inflated ranks with no error benefit; we corrected the policy, verified by direct decomposition that the saturated ranks (14, 29, 29) reach machine precision at every grid from 128^2 to 512^2 , and re-measured all results.

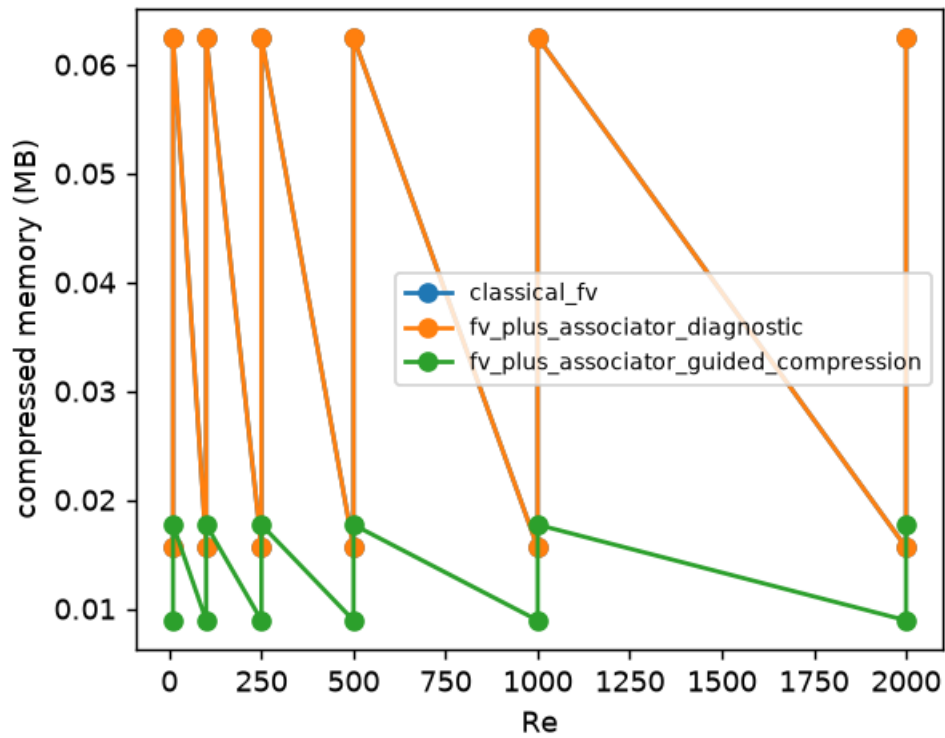
Perturbed-flow detail (Re=100, nx=64): 10% amplitude -> guided ranks (13, 26, 26), 11.1x at $8.0e-8$ error, 1.73x vs best uniform rank; 30% -> (11, 22, 22), 16.4x at $5.3e-6$, 1.67x. Perturbations are divergence-free multi-mode streamfunction fields.

B. Quantum Hardware Resource Estimates (measured ranks)

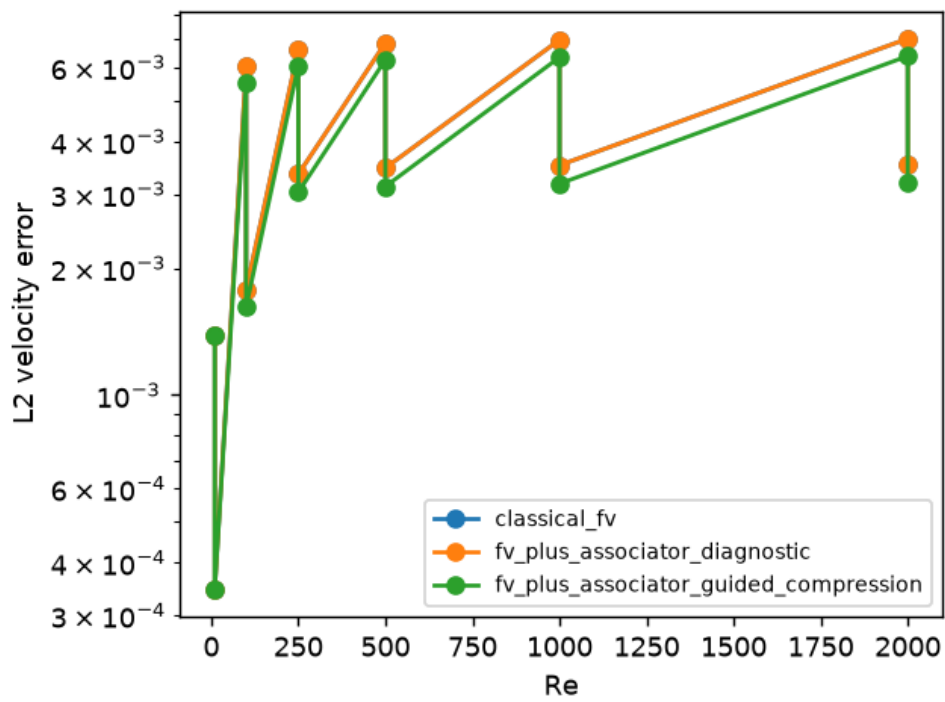
Grid	Ranks (t,x,y)	Logical qubits	Physical qubits*	Basis
64^2	(15, 30, 30)	~46	~46,000	measured
128^2	(14, 29, 29)	~50	~50,000	measured
256^2	(14, 29, 29)	~54	~54,000	measured
512^2	(14, 29, 29)	~57	~57,000	measured

*Surface code, ~1000 physical per logical qubit (d=15). State-preparation circuit depth: $O(2 \times 10^4)$ gates at 64^2 , $O(1.2 \times 10^5)$ at 128^2 . As nx doubles, spatial-mode qubit count grows by 1 while classical factor storage grows linearly - the logarithmic-vs-linear gap underlying the quantum memory pathway. Current devices (≤ 133 qubits) cannot execute these circuits; no hardware result is claimed.

C. Key Figures



Memory: dense classical vs compressed storage across the Reynolds sweep.



L2 velocity error vs Reynolds number for all three methods (32^2 , 64^2).